

Technische Hochschule Ingolstadt
Fakultät Elektro- und Informationstechnik

Sensor Network Technologies and Sensor Fusion

Simultaneous Localization and Mapping

EKF-SLAM

Dec 18, 2025

Welcome to the lecture!

Prof. Dr.-Ing. Björn Giesler

Technische Hochschule Ingolstadt

AIMotion, AI and Sensor Data Fusion

Email: bjoern.giesler@thi.de

- **Please register for the lecture on Moodle!**
  - Link: https://moodle.thi.de/course/view.php?id=10580
  - Key: **SensNT+DF#WS25**
- **News on Moodle:**
  - Slides for the lecture (upcoming lectures marked as WIP and are subject to change!)
  - Formulary, Exercises (Solutions provided after discussion in the lecture)
  - Exercises are optional, one task / sheet per week, but are highly recommended.
    They can be handed in for correction.
  - I will *try* to publish exercises before Wednesday‘s lecture so that initial questions can be discussed on Thursday.
- **Exam: Written exam, 90 minutes**
- **Consultation hours: Please arrange via e-mail to bjoern.giesler@thi.de.**
- **Please also don‘t hesitate to send questions, lecture feedback, change requests, etc.!**

Sensor Networks and Sensor Fusion

- **Previous lectures:**
  - Mathematical details of Particle Filters
  - Joint Probability Data Association Filters
  - JPDAs and Particle Filters

- **This week:**
  - Simultaneous Localization and Mapping
    - Introduction
    - EKF SLAM
    - Loop Closure
    - Particle Filter / Rao-Blackwellized Particle Filter SLAM
    - "Hidden Robot" problem
- **After the Christmas Break:**
  - Bit more SLAM
  - (DNN-based) early fusion approaches.



Merry Robot Christmas!

EKF-SLAM

EKF Recap

• Predict	$\hat{\mu}_{t+1\|t} = f(\hat{\mu}_{t\|t}, \mathbf{u}_t)$ $\Sigma_{t+1\|t} = F\Sigma_{t\|t}F^\top + Q$
• Measure	$\hat{\mathbf{z}}_{t+1} = H\hat{\mu}_{t+1\|t}$ $\mathbf{z}_{t+1} = h(\mu_{t+1})$
• Kalman Gain	$K_t = \Sigma_{t\|t-1}H^\top(H\Sigma_{t\|t-1}H^\top + R)^{-1}$
• Update Estimate	$\hat{\mu}_{t+1\|t+1} = \hat{\mu}_{t+1\|t} + K_t \cdot (\mathbf{z}_{t+1} - h(\hat{\mu}_{t+1\|t}))$
• Update Covariance	$\Sigma_{t+1\|t+1} = \Sigma_{t+1\|t} - K_t H \Sigma_{t+1\|t}$

with F the Jacobian of f and H the Jacobian of h.

We are adding a new term $\mathbf{u}_t$ for the control input, since we are now actively controlling and not just observing the modeled robot.

We are also changing notation to keep more in line with classical SLAM notation:

State $\mu = \mathbf{x}$

Covariance $\Sigma = P$

EKF Application to SLAM

- Estimate robot's pose and locations of landmarks in the environment with one filter
- State space (for a robot moving in 2D) is

$$\mu_t = (\underbrace{x_t, y_t, \theta_t}_{\text{robot pose } \mathbf{x}_t}, \underbrace{m_{1,x}, m_{1,y}}_{\text{landmark } \mathbf{m}_1}, \ldots, \underbrace{m_{n,x}, m_{n,y}}_{\text{landmark } \mathbf{m}_n})^\top$$

- Covariance matrix models joint uncertainties between robot pose and individual landmarks:

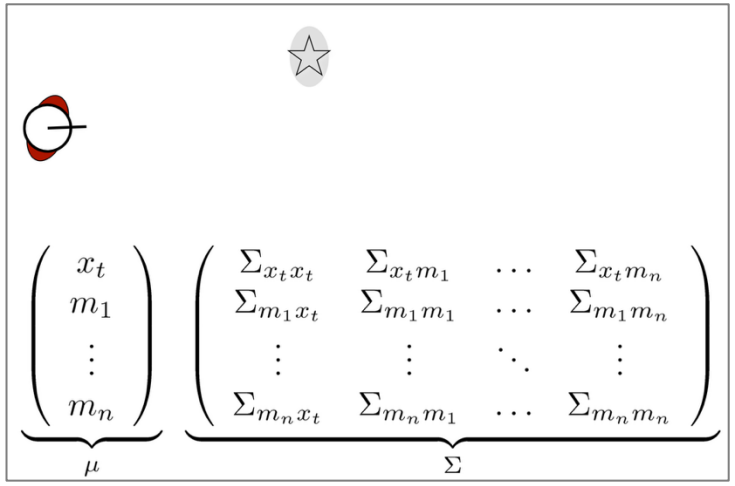
$$\Sigma = \begin{bmatrix} \Sigma_{\mathbf{x}_t\mathbf{x}_t} & \Sigma_{\mathbf{x}_t\mathbf{m}_1} & \cdots & \Sigma_{\mathbf{x}_t\mathbf{m}_n} \\ \Sigma_{\mathbf{m}_1\mathbf{x}_t} & \Sigma_{\mathbf{m}_1\mathbf{m}_1} & \cdots & \Sigma_{\mathbf{m}_1\mathbf{m}_n} \\ \vdots & \vdots & \ddots & \vdots \\ \Sigma_{\mathbf{m}_n\mathbf{x}_t} & \Sigma_{\mathbf{m}_n\mathbf{m}_1} & \cdots & \Sigma_{\mathbf{m}_n\mathbf{m}_n} \end{bmatrix}$$

or more compact:

$$\mathbf{s} = \begin{bmatrix} \mathbf{x} \\ \mathbf{m} \end{bmatrix}$$

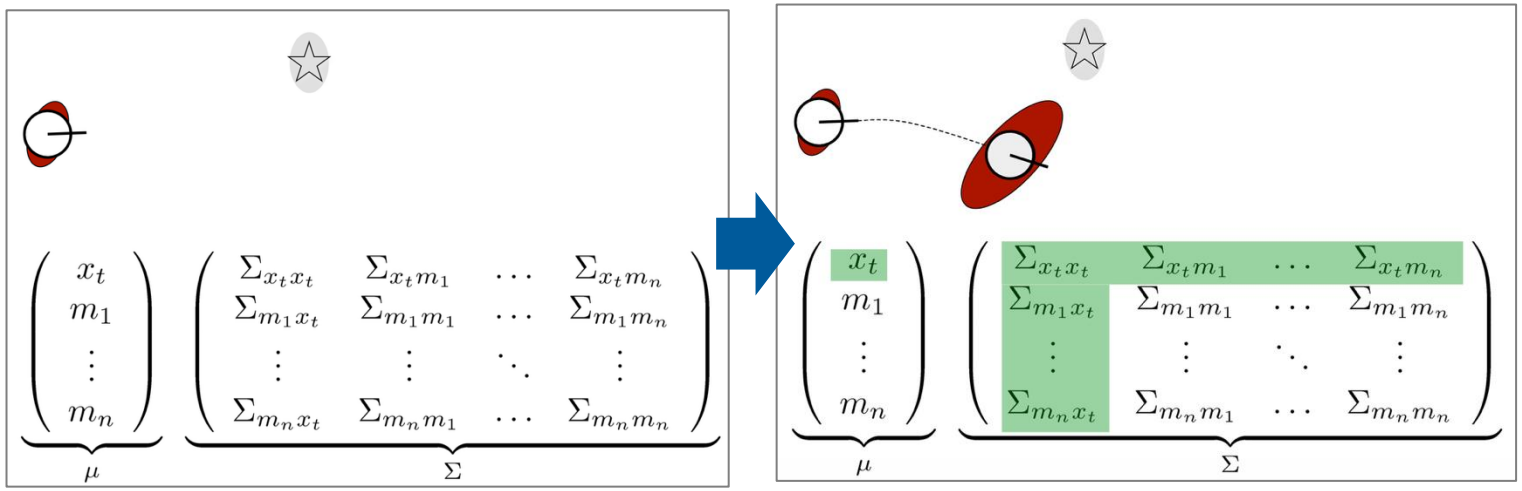
$$\Sigma = \begin{bmatrix} \Sigma_{\mathbf{xx}} & \Sigma_{\mathbf{xm}} \\ \Sigma_{\mathbf{mx}} & \Sigma_{\mathbf{mm}} \end{bmatrix}$$

Initial State

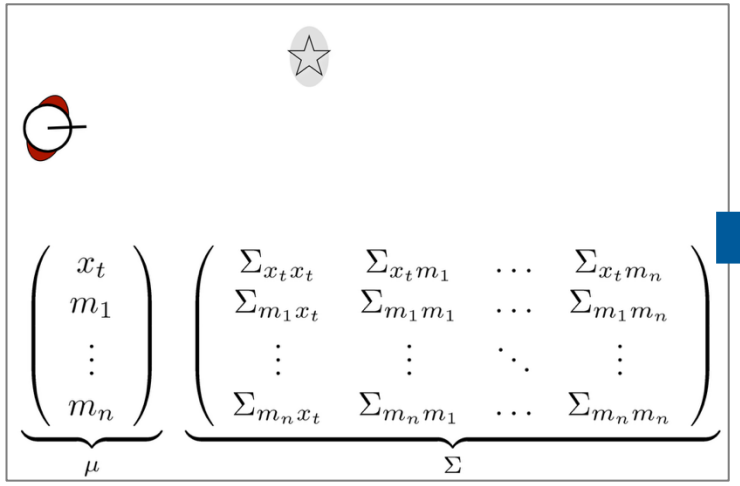


Initial State

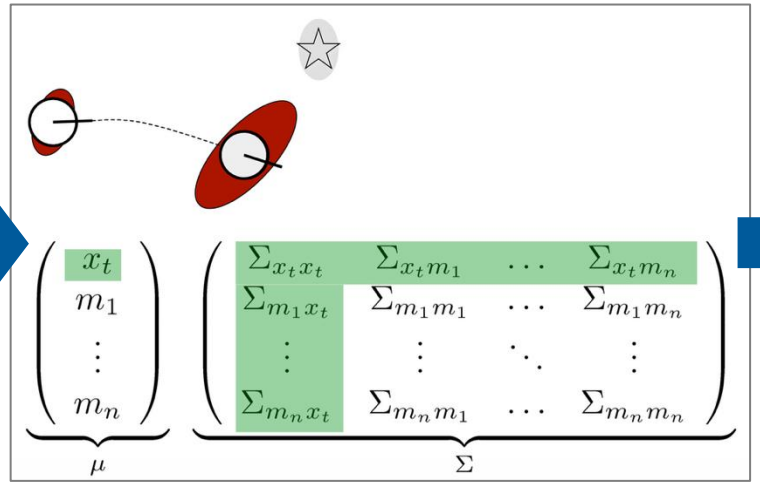
Predicted Motion $\hat{\mathbf{x}}_t = f(\hat{\mathbf{x}}_{t-1}, \mathbf{u}_t)$



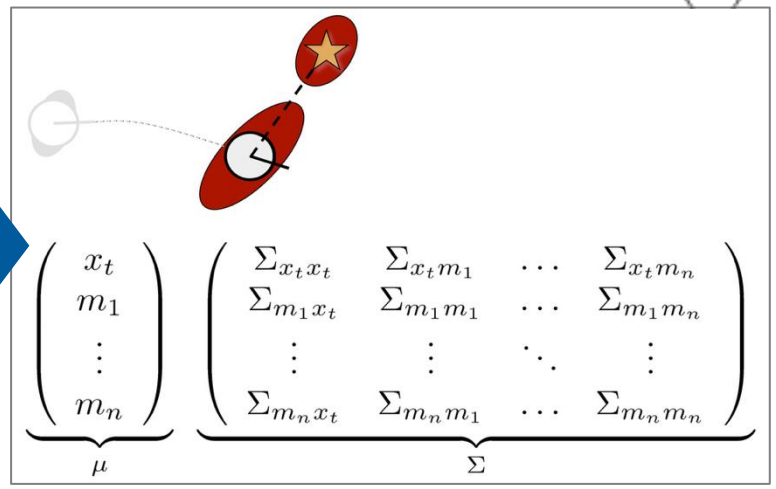
Initial State



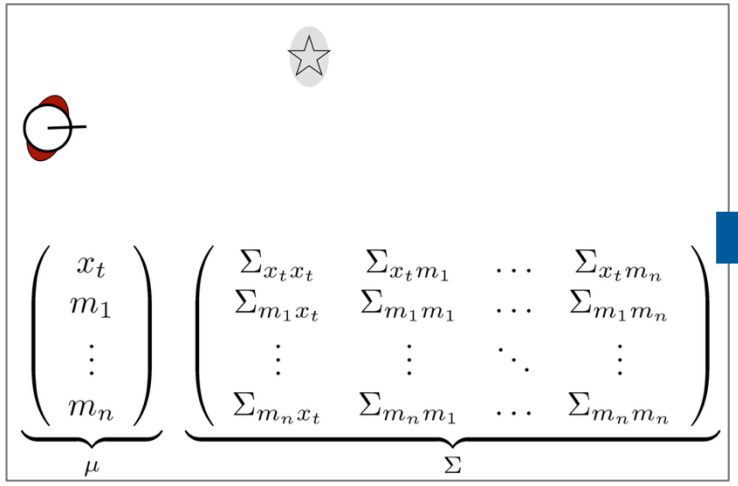
Predicted Motion $\hat{\mathbf{x}}_t = f(\hat{\mathbf{x}}_{t-1}, \mathbf{u}_t)$



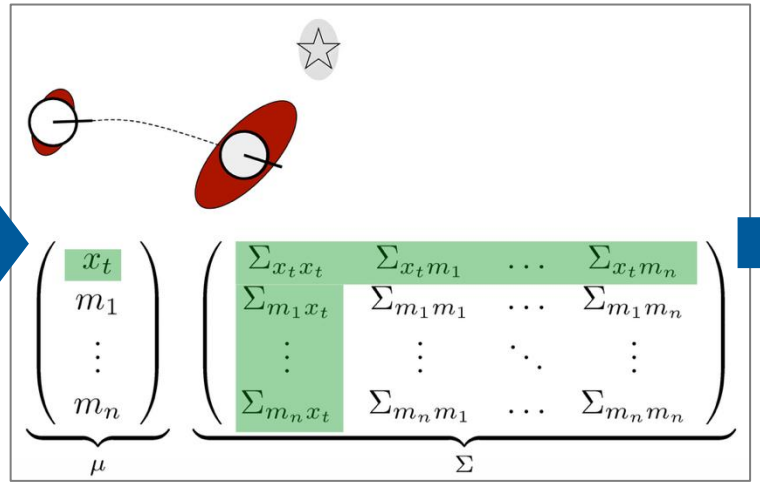
Predicted Measurement $\hat{\mathbf{z}} = h(\hat{\mathbf{x}})$



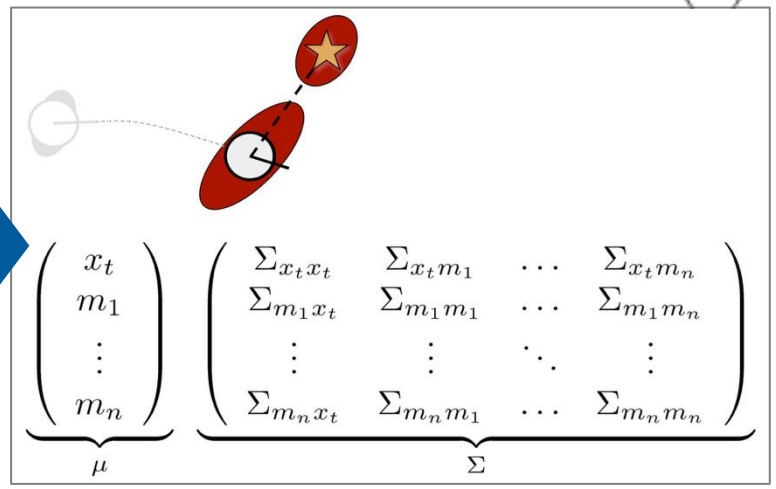
Initial State



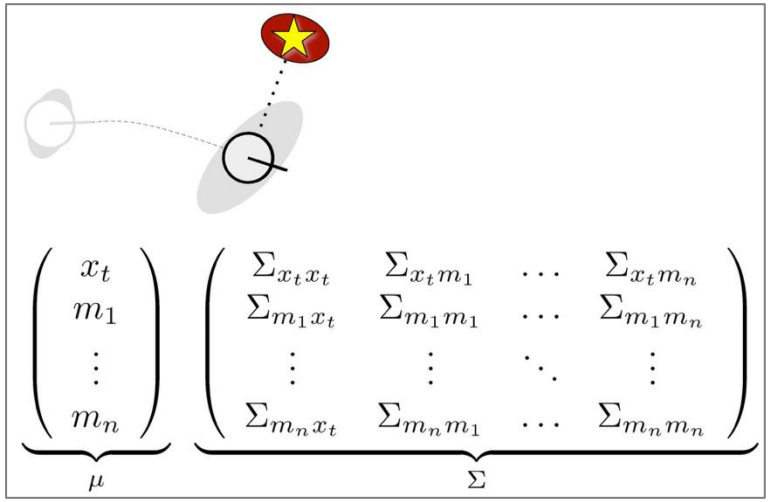
Predicted Motion $\hat{\mathbf{x}}_t = f(\hat{\mathbf{x}}_{t-1}, \mathbf{u}_t)$



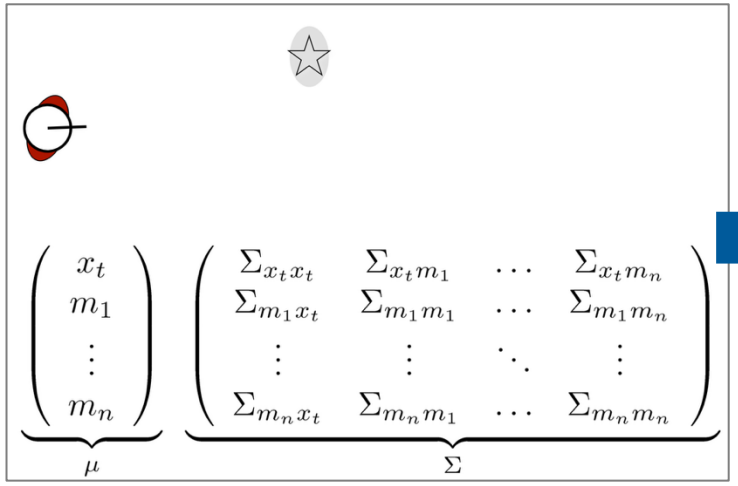
Predicted Measurement $\hat{\mathbf{z}} = h(\hat{\mathbf{x}})$



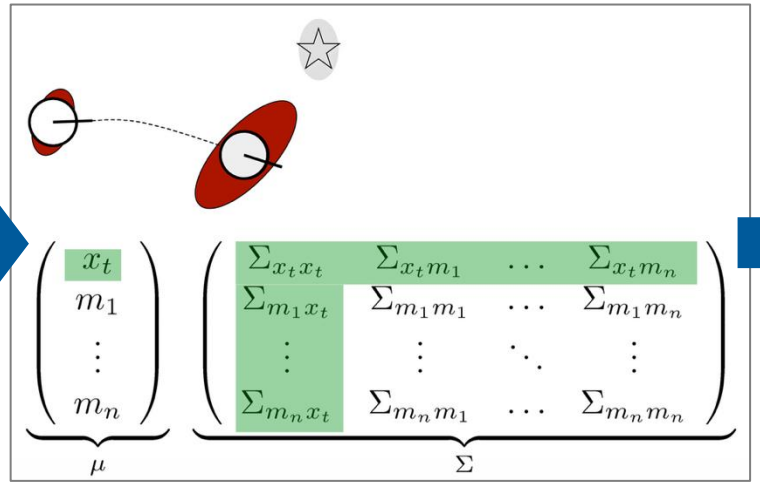
Obtained Measurement $\mathbf{z}$



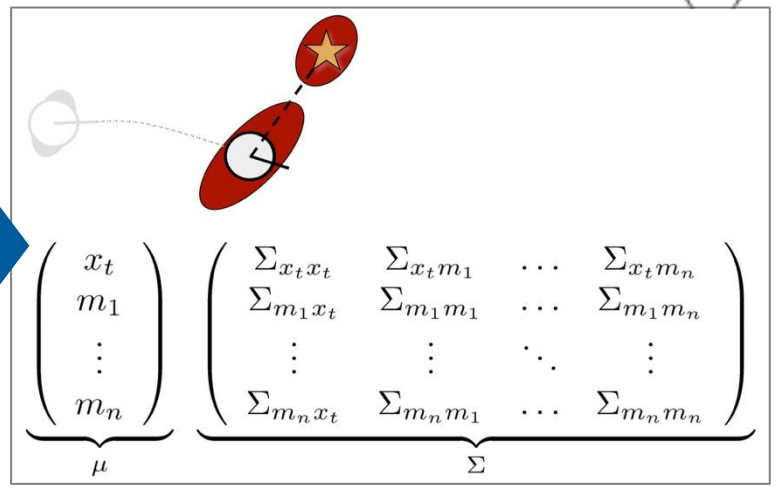
Initial State



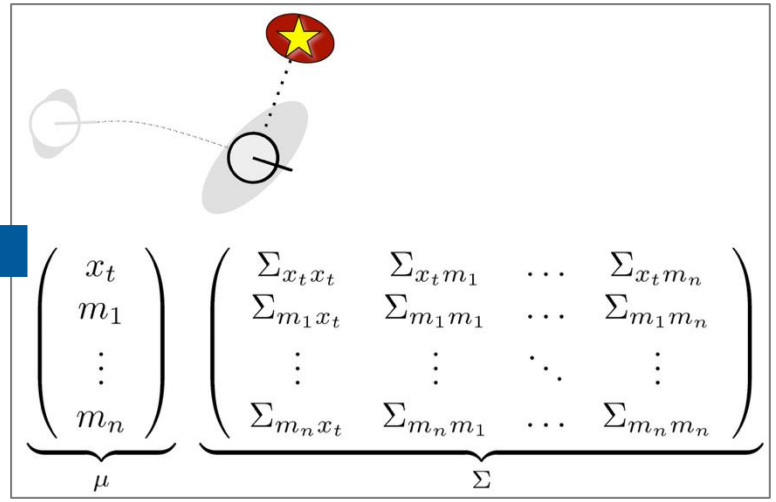
Predicted Motion $\hat{\mathbf{x}}_t = f(\hat{\mathbf{x}}_{t-1}, \mathbf{u}_t)$



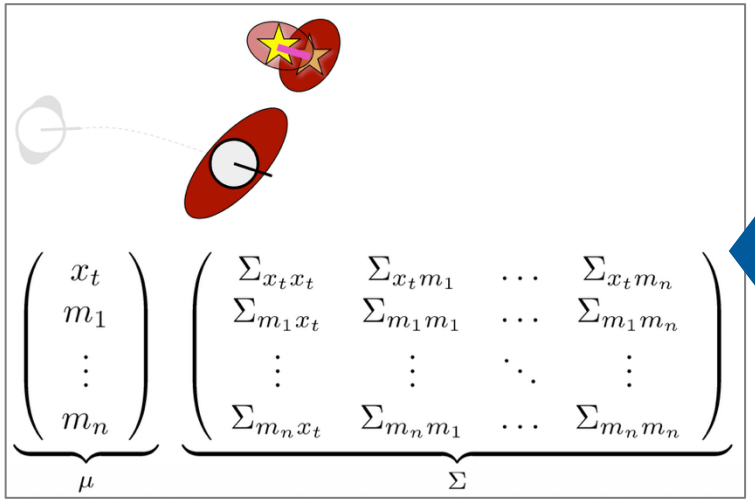
Predicted Measurement $\hat{\mathbf{z}} = h(\hat{\mathbf{x}})$

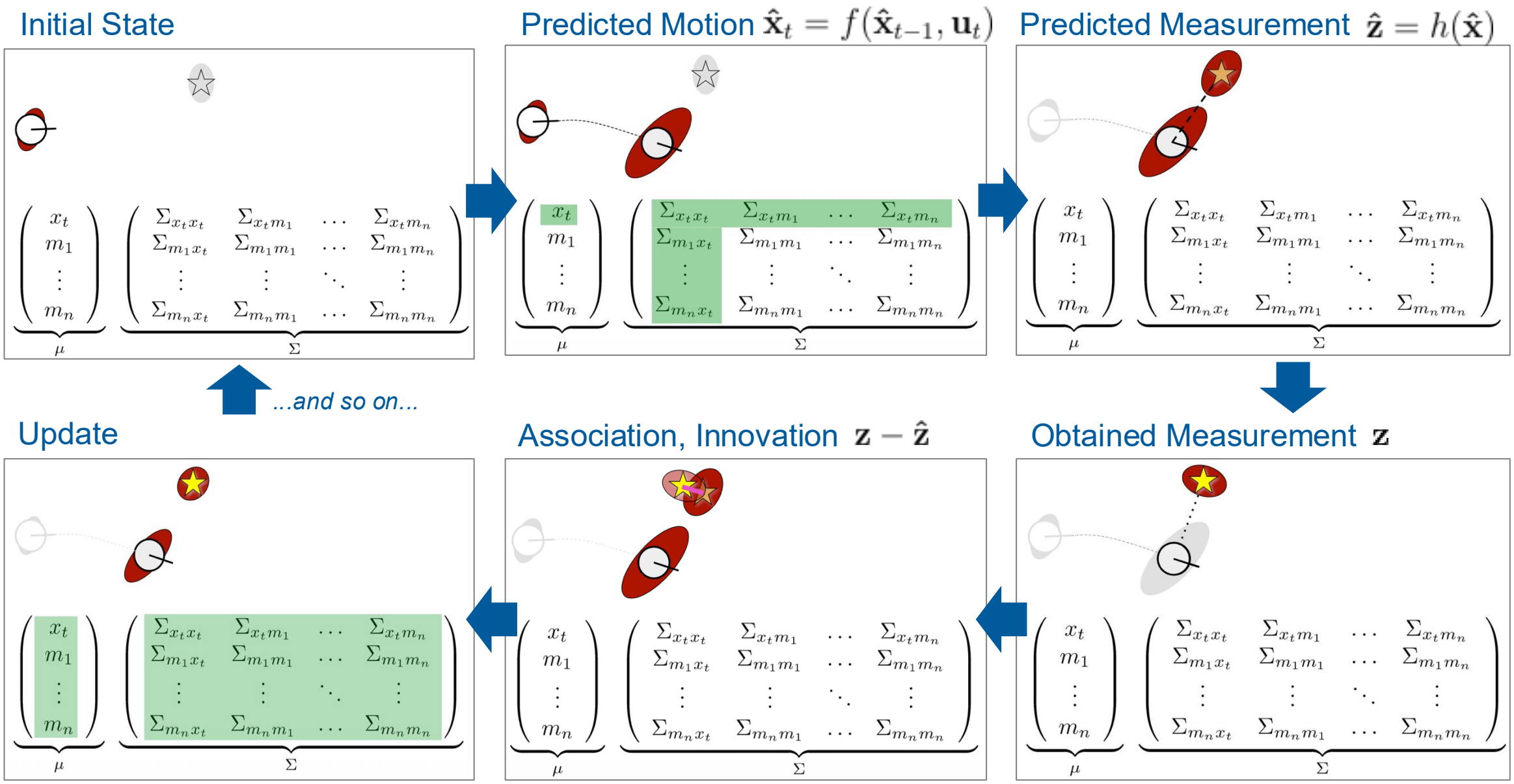


Obtained Measurement $\mathbf{z}$



Association, Innovation $\mathbf{z} - \hat{\mathbf{z}}$




Initial State
Predicted Motion $\hat{\mathbf{x}}_t = f(\hat{\mathbf{x}}_{t-1}, \mathbf{u}_t)$
Predicted Measurement $\hat{\mathbf{z}} = h(\hat{\mathbf{x}})$
$\begin{pmatrix} x_t \\ m_1 \\ \vdots \\ m_n \end{pmatrix}$ μ
$\begin{pmatrix} \Sigma_{x_t x_t} & \Sigma_{x_t m_1} & \cdots & \Sigma_{x_t m_n} \\ \Sigma_{m_1 x_t} & \Sigma_{m_1 m_1} & \cdots & \Sigma_{m_1 m_n} \\ \vdots & \vdots & \ddots & \vdots \\ \Sigma_{m_n x_t} & \Sigma_{m_n m_1} & \cdots & \Sigma_{m_n m_n} \end{pmatrix}$ Σ
...and so on...
Update
Association, Innovation $\mathbf{z} - \hat{\mathbf{z}}$
Obtained Measurement $\mathbf{z}$

EKF-SLAM: Concrete Example

Setup

- Mobile robot that moves in the 2D plane

  $$\mathbf{x} = (x, y, \theta)^\top$$

- Velocity-based motion model as input

  Forward velocity v

  Turn rate ω

  $$\mathbf{u} = (v, \omega)^\top$$

- Observation of point landmarks
- Sensor delivering range and azimuth (bearing)
- Known data association
- Known number of landmarks

EKF Algorithm:

$$\hat{\mu}_{t+1|t} = f(\hat{\mu}_{t|t}, \mathbf{u}_t)$$

$$\Sigma_{t+1|t} = F\Sigma_{t|t}F^\top + Q$$

$$K_t = \Sigma_{t|t-1}H^\top(H\Sigma_{t|t-1}H^\top + R)^{-1}$$

$$\hat{\mu}_{t+1|t+1} = \hat{\mu}_{t+1|t} + K_t \cdot (\mathbf{z}_{t+1} - H\hat{\mu}_{t+1|t})$$

$$\Sigma_{t+1|t+1} = \Sigma_{t+1|t} - K_t H \Sigma_{t+1|t}$$

Initialization

- **Platform starts in its own reference frame, at** $(0,0,0)$
- **All landmarks are unknown**
- **Size of state space:** $2N + 3$

$$\mu_0 = (0, 0, 0, \ldots 0)^\top$$

$$\Sigma_0 = \begin{bmatrix} 0 & 0 & 0 & \ldots & 0 \\ 0 & 0 & 0 & \ldots & 0 \\ 0 & 0 & 0 & \ldots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \ldots & 0 \end{bmatrix}$$

Prediction Step: Robot Motion

- Motion in the plane is

$$\begin{bmatrix} x' \\ y' \\ \theta' \end{bmatrix} = \underbrace{\begin{bmatrix} x \\ y \\ \theta \end{bmatrix} + \begin{bmatrix} -\frac{v_t}{\omega_t}\sin\theta + \frac{v_t}{\omega_t}\sin(\theta+\omega_t\Delta t) \\ \frac{v_t}{\omega_t}\cos\theta - \frac{v_t}{\omega_t}\cos(\theta+\omega_t\Delta t) \\ \omega_t\Delta t \end{bmatrix}}_{f((x,y,\theta)^\top,\mathbf{u}_t)}$$

- We map this to the 2N+3-dimensional space with

$$\begin{bmatrix} x' \\ y' \\ \theta' \\ \vdots \end{bmatrix} = \underbrace{\begin{bmatrix} x \\ y \\ \theta \\ \vdots \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ \vdots & \vdots & \vdots \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} -\frac{v_t}{\omega_t}\sin\theta + \frac{v_t}{\omega_t}\sin(\theta+\omega_t\Delta t) \\ \frac{v_t}{\omega_t}\cos\theta - \frac{v_t}{\omega_t}\cos(\theta+\omega_t\Delta t) \\ \omega_t\Delta t \end{bmatrix}}_{f(\mu_t,\mathbf{u}_t)}$$

$$\hat{\mu}_{t+1|t} = f(\hat{\mu}_{t|t}, \mathbf{u}_t)$$

$$\Sigma_{t+1|t} = F\Sigma_{t|t}F^\top + Q$$

$$K_t = \Sigma_{t|t-1}H^\top(H\Sigma_{t|t-1}H^\top + R)^{-1}$$

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$$\Sigma_{t+1|t+1} = \Sigma_{t+1|t} - K_t H \Sigma_{t+1|t}$$

As we can see, $f(\cdot)$ only affects the motion and not the landmarks.

Deriving the Jacobian of the motion

Since we only touch the first 3 components of the state, we can write F as a composition of a 3x3 matrix and the $2Nx2N$ identity matrix.

$$F_t = \begin{bmatrix} F_t^x & 0 \\ 0 & I \end{bmatrix}$$

$$\hat{\mu}_{t+1|t} = f(\hat{\mu}_{t|t}, \mathbf{u}_t)$$

$$\Sigma_{t+1|t} = F\Sigma_{t|t}F^\top + Q$$

$$K_t = \Sigma_{t|t-1}H^\top(H\Sigma_{t|t-1}H^\top + R)^{-1}$$

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Deriving the Jacobian of the motion

$$F_t^x = \frac{\partial}{\partial(x,y,\theta)^\top}\left[\begin{pmatrix} x \\ y \\ \theta \end{pmatrix} + \begin{pmatrix} -\frac{v_t}{\omega_t}\sin\theta + \frac{v_t}{\omega_t}\sin(\theta+\omega_t\Delta t) \\ \frac{v_t}{\omega_t}\cos\theta - \frac{v_t}{\omega_t}\cos(\theta+\omega_t\Delta t) \\ \omega_t\Delta t \end{pmatrix}\right]$$

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Deriving the Jacobian of the motion

$$F_t^x = \frac{\partial}{\partial(x,y,\theta)^\top}\left[\begin{pmatrix} x \\ y \\ \theta \end{pmatrix} + \begin{pmatrix} -\frac{v_t}{\omega_t}\sin\theta + \frac{v_t}{\omega_t}\sin(\theta+\omega_t\Delta t) \\ \frac{v_t}{\omega_t}\cos\theta - \frac{v_t}{\omega_t}\cos(\theta+\omega_t\Delta t) \\ \omega_t\Delta t \end{pmatrix}\right]$$

$$= I + \frac{\partial}{\partial(x,y,\theta)^\top}\begin{pmatrix} -\frac{v_t}{\omega_t}\sin\theta + \frac{v_t}{\omega_t}\sin(\theta+\omega_t\Delta t) \\ \frac{v_t}{\omega_t}\cos\theta - \frac{v_t}{\omega_t}\cos(\theta+\omega_t\Delta t) \\ \omega_t\Delta t \end{pmatrix}$$

$$\hat{\mu}_{t+1|t} = f(\hat{\mu}_{t|t}, \mathbf{u}_t)$$

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Deriving the Jacobian of the motion

$$
\begin{aligned}
F_t^x &= \frac{\partial}{\partial (x,y,\theta)^\top} \left[\begin{pmatrix} x \\ y \\ \theta \end{pmatrix} + \begin{pmatrix} -\frac{v_t}{\omega_t}\sin\theta + \frac{v_t}{\omega_t}\sin(\theta+\omega_t\Delta t) \\ \frac{v_t}{\omega_t}\cos\theta - \frac{v_t}{\omega_t}\cos(\theta+\omega_t\Delta t) \\ \omega_t \Delta t \end{pmatrix} \right] \\
&= I + \frac{\partial}{\partial (x,y,\theta)^\top} \begin{pmatrix} -\frac{v_t}{\omega_t}\sin\theta + \frac{v_t}{\omega_t}\sin(\theta+\omega_t\Delta t) \\ \frac{v_t}{\omega_t}\cos\theta - \frac{v_t}{\omega_t}\cos(\theta+\omega_t\Delta t) \\ \omega_t \Delta t \end{pmatrix} \\
&= I + \begin{pmatrix} 0 & 0 & -\frac{v_t}{\omega_t}\cos\theta + \frac{v_t}{\omega_t}\cos(\theta+\omega_t\Delta t) \\ 0 & 0 & -\frac{v_t}{\omega_t}\sin\theta + \frac{v_t}{\omega_t}\sin(\theta+\omega_t\Delta t) \\ 0 & 0 & 0 \end{pmatrix}
\end{aligned}
$$

$$\hat{\mu}_{t+1|t} = f(\hat{\mu}_{t|t}, \mathbf{u}_t)$$

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Deriving the Jacobian of the motion

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\begin{aligned}
F_t^x &= \frac{\partial}{\partial (x,y,\theta)^\top} \left[\begin{pmatrix} x \\ y \\ \theta \end{pmatrix} + \begin{pmatrix} -\frac{v_t}{\omega_t}\sin\theta + \frac{v_t}{\omega_t}\sin(\theta+\omega_t\Delta t) \\ \frac{v_t}{\omega_t}\cos\theta - \frac{v_t}{\omega_t}\cos(\theta+\omega_t\Delta t) \\ \omega_t \Delta t \end{pmatrix} \right] \\
&= I + \frac{\partial}{\partial (x,y,\theta)^\top} \begin{pmatrix} -\frac{v_t}{\omega_t}\sin\theta + \frac{v_t}{\omega_t}\sin(\theta+\omega_t\Delta t) \\ \frac{v_t}{\omega_t}\cos\theta - \frac{v_t}{\omega_t}\cos(\theta+\omega_t\Delta t) \\ \omega_t \Delta t \end{pmatrix} \\
&= I + \begin{pmatrix} 0 & 0 & -\frac{v_t}{\omega_t}\cos\theta + \frac{v_t}{\omega_t}\cos(\theta+\omega_t\Delta t) \\ 0 & 0 & -\frac{v_t}{\omega_t}\sin\theta + \frac{v_t}{\omega_t}\sin(\theta+\omega_t\Delta t) \\ 0 & 0 & 0 \end{pmatrix} \\
&= \begin{pmatrix} 1 & 0 & \frac{v_t}{\omega_t}(\cos(\theta+\omega_t\Delta t) - \cos\theta) \\ 0 & 1 & \frac{v_t}{\omega_t}(\sin(\theta+\omega_t\Delta t) - \sin\theta) \\ 0 & 0 & 1 \end{pmatrix}
\end{aligned}
$$

$$\hat{\mu}_{t+1|t} = f(\hat{\mu}_{t|t}, \mathbf{u}_t)$$

$$\Sigma_{t+1|t} = F\Sigma_{t|t}F^\top + Q$$

$$K_t = \Sigma_{t|t-1}H^\top(H\Sigma_{t|t-1}H^\top + R)^{-1}$$

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$$\Sigma_{t+1|t+1} = \Sigma_{t+1|t} - K_t H \Sigma_{t+1|t}$$

Prediction Equations Complete!

$$\Sigma_{t+1|t} = F\Sigma_{t|t}F^\top + Q$$
$$= \begin{pmatrix} F_t^x & 0 \\ 0 & I \end{pmatrix} \begin{pmatrix} \Sigma_{xx} & \Sigma_{xm} \\ \Sigma_{mx} & \Sigma_{mm} \end{pmatrix} \begin{pmatrix} (F_t^x)^\top & 0 \\ 0 & I \end{pmatrix} + Q$$
$$= \begin{pmatrix} F_t^x \Sigma xx (F_t^x)^\top & F_t^x \Sigma_{xm} \\ (F_t^x \Sigma_{xm})^\top & \Sigma_{mm} \end{pmatrix} + Q$$

$$\hat{\mu}_{t+1|t} = f(\hat{\mu}_{t|t}, \mathbf{u}_t)$$
$$\Sigma_{t+1|t} = F\Sigma_{t|t}F^\top + Q$$

$$K_t = \Sigma_{t|t-1}H^\top(H\Sigma_{t|t-1}H^\top + R)^{-1}$$

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$$\Sigma_{t+1|t+1} = \Sigma_{t+1|t} - K_t H\Sigma_{t+1|t}$$

EKF-SLAM: Correction Step

- We assume a known (and correct ☺) data association.
- i-th measurement at time t observes the landmark with index j:

$$c_t^i = j$$

- If the landmark was unobserved previously, initialize it
- Compute the expected observation
- Compute the Jacobian of h
- Compute the Kalman gain

$$\hat{\mu}_{t+1|t} = f(\hat{\mu}_{t|t}, \mathbf{u}_t)$$
$$\Sigma_{t+1|t} = F\Sigma_{t|t}F^\top + Q$$

$$K_t = \Sigma_{t|t-1}H^\top(H\Sigma_{t|t-1}H^\top + R)^{-1}$$
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$$\Sigma_{t+1|t+1} = \Sigma_{t+1|t} - K_t H \Sigma_{t+1|t}$$

Range-Azimuth Observation

- **We obtain i observations in ray space, i.e. a distance range and an azimuth angle:** $z_t^i = (r_t^i, \phi_t^i)^\top$.
- **The location in cartesian coordinates can be expressed by:**

$$\begin{pmatrix} \hat{\mu}_{j,x} \\ \hat{\mu}_{j,y} \end{pmatrix} = \begin{pmatrix} \hat{\mu}_{t,x} \\ \hat{\mu}_{t,y} \end{pmatrix} + \begin{pmatrix} r_t^i \cos(\phi_t^i + \hat{\mu}_{t,\theta} \\ r_t^i \sin(\phi_t^i + \hat{\mu}_{t,\theta} \end{pmatrix}.$$

location of landmark j — estimated robot location — relative measurement

- **If the landmark was previously unobserved, we initialize it with the above equation.**
- **We can now compute the expected observation:**

$$\begin{pmatrix} \delta_x \\ \delta_y \end{pmatrix} = \begin{pmatrix} \hat{\mu}_{j,x} - \hat{\mu}_{t,x} \\ \hat{\mu}_{j,y} - \hat{\mu}_{t,y} \end{pmatrix}$$
$$\hat{\mathbf{z}}_t^i = h(\hat{\mu}_t) = \begin{pmatrix} \sqrt{\delta_x^2 + \delta_y^2} \\ \arctan\frac{\delta_y}{\delta_x} - \hat{\mu}_{t,\theta} \end{pmatrix}$$

Jacobian for the Observation

- Compute the Jacobian for $\hat{\mathbf{z}}_t^i = h(\hat{\mu}_t) = \begin{pmatrix} \sqrt{\delta_x^2 + \delta_y^2} \\ \arctan \frac{\delta_y}{\delta_x} - \hat{\mu}_{t,\theta} \end{pmatrix}$.

- This is again not touching the entire state vector, but only the motion state and the j-th landmark:

$$(x, y, \theta, m_{j,x}, m_{j_y})^\top$$

- Therefore we only need to compute a low-dimensional "excerpt" of H: $H_{t,\text{low}}^i = \frac{\partial h(\hat{\mu}_t)}{\partial \mu_t}$

- For brevity, we introduce $q = \delta_x^2 + \delta_y^2$.

$$\frac{\partial h_r}{\partial x} = \frac{\partial \sqrt{q}}{\partial x} = -\frac{1}{2\sqrt{q}} \cdot 2\delta_x = -\frac{\delta_x}{\sqrt{q}}, \quad \frac{\partial h_r}{\partial y} = -\frac{\delta_y}{\sqrt{q}}, \quad \frac{\partial h_r}{\partial \theta} = 0, \quad \frac{\partial h_r}{\partial m_{j,x}} = \frac{\delta_x}{\sqrt{q}}, \quad \frac{\partial h_r}{\partial m_{j,y}} = \frac{\delta_y}{\sqrt{q}}$$

$$\frac{\partial h_\theta}{\partial x} = \frac{\partial \arctan \frac{\delta_y}{\delta_x}}{\partial x} = \frac{1}{1 + (\frac{\delta_y}{\delta_x})^2} \cdot \frac{\delta_y}{\delta_x^2} = \frac{\delta_x^2}{\delta_x^2 + \delta_y^2} \cdot \frac{\delta_y}{\delta_x^2} = \frac{\delta_y}{q},$$

$$\frac{\partial h_\theta}{\partial y} = -\frac{\delta_x}{q}, \quad \frac{\partial h_\theta}{\partial \theta} = -1, \quad \frac{\partial h_\theta}{\partial m_{j,x}} = -\frac{\delta_y}{q}, \quad \frac{\partial h_\theta}{\partial m_{j,y}} = \frac{\delta_x}{q}.$$

Jacobian for the Observation

- With the above, we can write

$$H^i_{t,\text{low}} = \frac{1}{q}\begin{pmatrix} -\sqrt{q}\delta_x & -\sqrt{q}\delta_y & 0 & \sqrt{q}\delta_x & \sqrt{q}\delta_y \\ \delta_y & -\delta_x & -q & -\delta_y & \delta_x \end{pmatrix}.$$

- We can transfer it to high-dimensional space by multiplying with a mapping matrix $M_{x,j}$:

$$M_{x,j} = \begin{pmatrix} 1 & 0 & 0 & \underbrace{0 \ldots 0}_{2j-2} & 0 & 0 & \underbrace{0 \ldots 0}_{2N-2j} \\ 0 & 1 & 0 & 0 \ldots 0 & 0 & 0 & 0 \ldots 0 \\ 0 & 0 & 1 & 0 \ldots 0 & 0 & 0 & 0 \ldots 0 \\ 0 & 0 & 1 & 0 \ldots 0 & 1 & 0 & 0 \ldots 0 \\ 0 & 0 & 1 & 0 \ldots 0 & 0 & 1 & 0 \ldots 0 \end{pmatrix}$$

$$H^i_t = H^i_{t,\text{low}} M_{x,j}.$$

$$\hat{\mu}_{t+1|t} = f(\hat{\mu}_{t|t}, \mathbf{u}_t)$$

$$\Sigma_{t+1|t} = F\Sigma_{t|t}F^\top + Q$$

$$K_t = \Sigma_{t|t-1}H^\top(H\Sigma_{t|t-1}H^\top + R)^{-1}$$

$$\hat{\mu}_{t+1|t+1} = \hat{\mu}_{t+1|t} + K_t \cdot (\mathbf{z}_{t+1} - H\hat{\mu}_{t+1|t})$$

$$\Sigma_{t+1|t+1} = \Sigma_{t+1|t} - K_t H \Sigma_{t+1|t}$$

Full Correction Step

1: **for** all observed features $z_t^i = (r_t^i, \phi_t^i)$ **do**

2: $\quad j = c_t^i$

3: $\quad$ **if** landmark j never seen before **then**

4: $$\begin{pmatrix} \hat{\mu}_{j,x} \\ \hat{\mu}_{j,y} \end{pmatrix} = \begin{pmatrix} \hat{\mu}_{t,x} \\ \hat{\mu}_{t,y} \end{pmatrix} + \begin{pmatrix} r_t^i \cos(\phi_t^i + \hat{\mu}_{t,\theta} \\ r_t^i \sin(\phi_t^i + \hat{\mu}_{t,\theta} \end{pmatrix}$$

5: $\quad$ **end if**

6: $$\begin{pmatrix} \delta_x \\ \delta_y \end{pmatrix} = \begin{pmatrix} \hat{\mu}_{j,x} - \hat{\mu}_{t,x} \\ \hat{\mu}_{j,y} - \hat{\mu}t, y \end{pmatrix}$$

7: $q = \delta_x^2 + \delta_y^2$

8: $$\hat{\mathbf{z}}_t^i = \begin{pmatrix} \sqrt{q} \\ \arctan \frac{\delta_y}{\delta_x} - \hat{\mu}_{t,\theta} \end{pmatrix}$$

9: $M_{x,j} = \ldots$

10: $$H_t^i = \frac{1}{q} \begin{pmatrix} -\sqrt{q}\delta_x & -\sqrt{q}\delta_y & 0 & \sqrt{q}\delta_x & \sqrt{q}\delta_y \\ \delta_y & -\delta_x & -q & -\delta_y & \delta_x \end{pmatrix} M_{x,j}$$

11: $\quad$ Update K, $\hat{\mu}_t$, Σ_t

12: **end for**

$$\hat{\mu}_{t+1|t} = f(\hat{\mu}_{t|t}, \mathbf{u}_t)$$

$$\Sigma_{t+1|t} = F\Sigma_{t|t}F^\top + Q$$

$$K_t = \Sigma_{t|t-1}H^\top(H\Sigma_{t|t-1}H^\top + R)^{-1}$$

$$\hat{\mu}_{t+1|t+1} = \hat{\mu}_{t+1|t} + K_t \cdot (\mathbf{z}_{t+1} - H\hat{\mu}_{t+1|t})$$

$$\Sigma_{t+1|t+1} = \Sigma_{t+1|t} - K_t H \Sigma_{t+1|t}$$

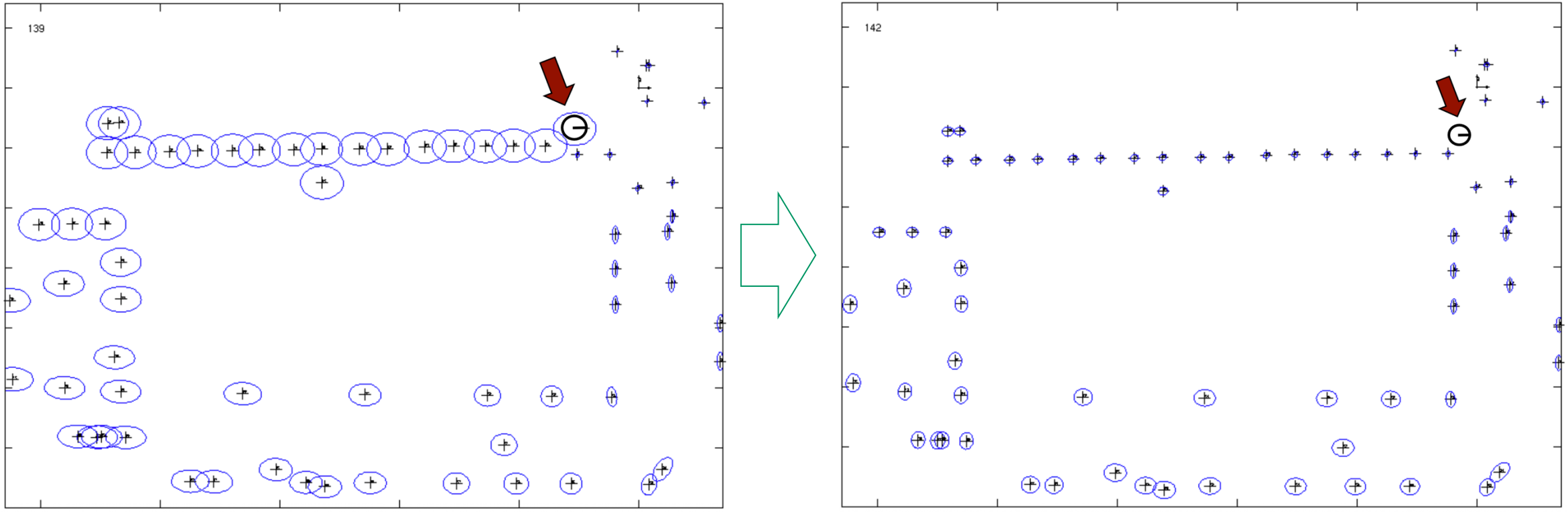
Done!

Loop Closing and Correlations

Loop Closing

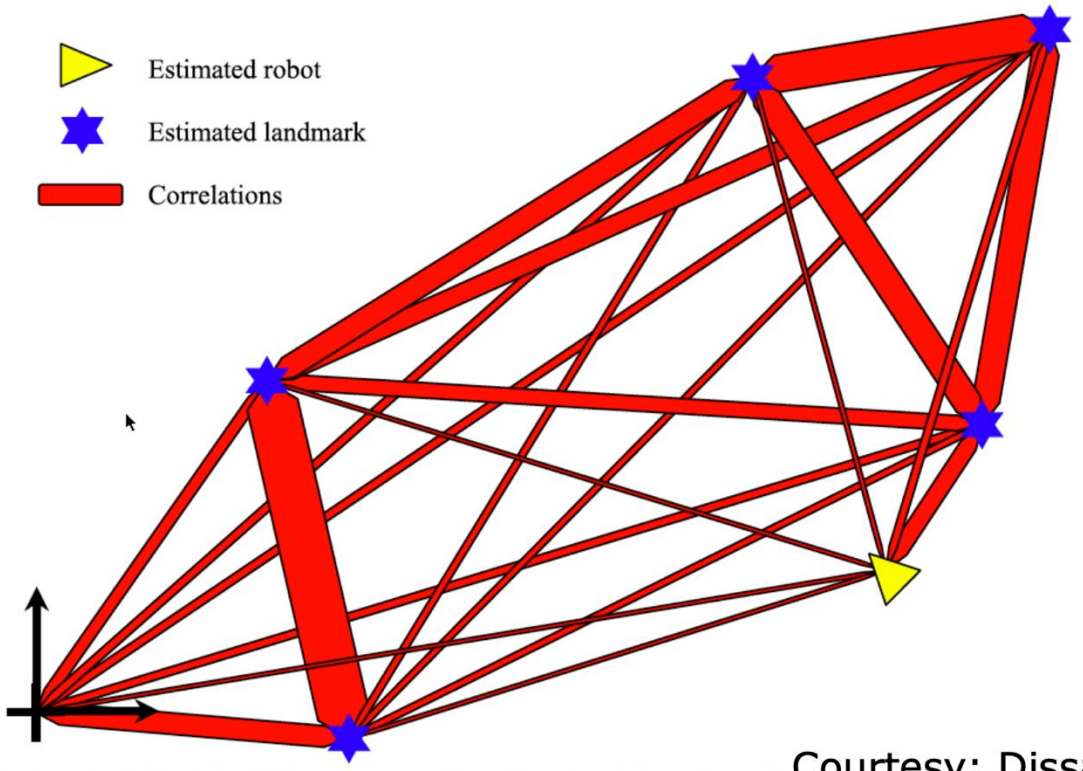
- Loop closing means revisiting (and recognizing!) an already mapped area
- Data association under high ambiguity and possible environment symmetries...
- Uncertainties collapse after a loop closure (whether the closure was correct or not...)
- Loop closing reduces the uncertainty in robot and landmark estimates.
- This can be exploited when exploring an environment for the sake of better (e.g. more accurate) maps.
- However, wrong loop closures lead to filter divergence.

Loop Closing



EKF-SLAM Correlations

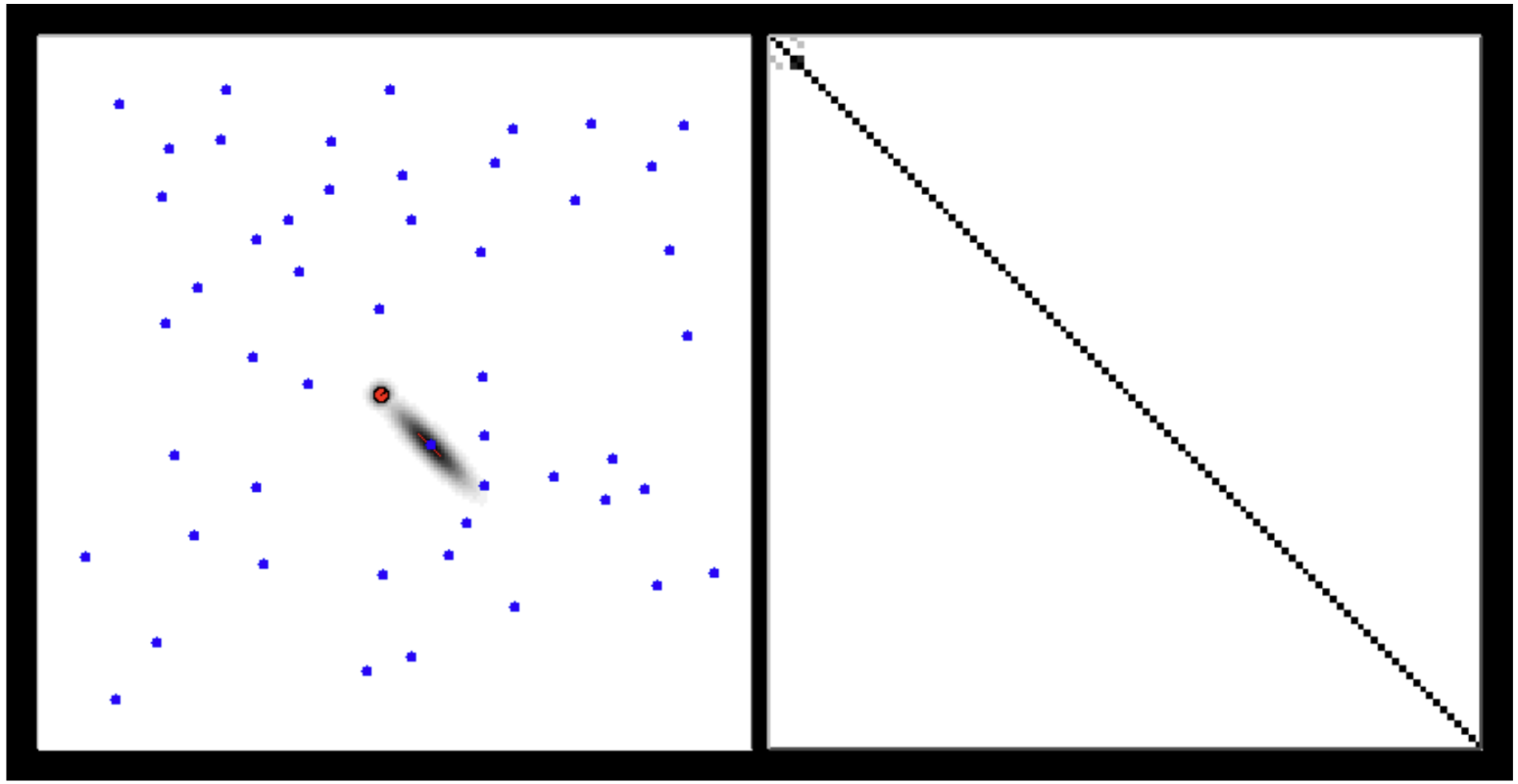
- In the limit, the landmark estimates become fully correlated.



Courtesy: Dissanayake

EKF-SLAM Correlations

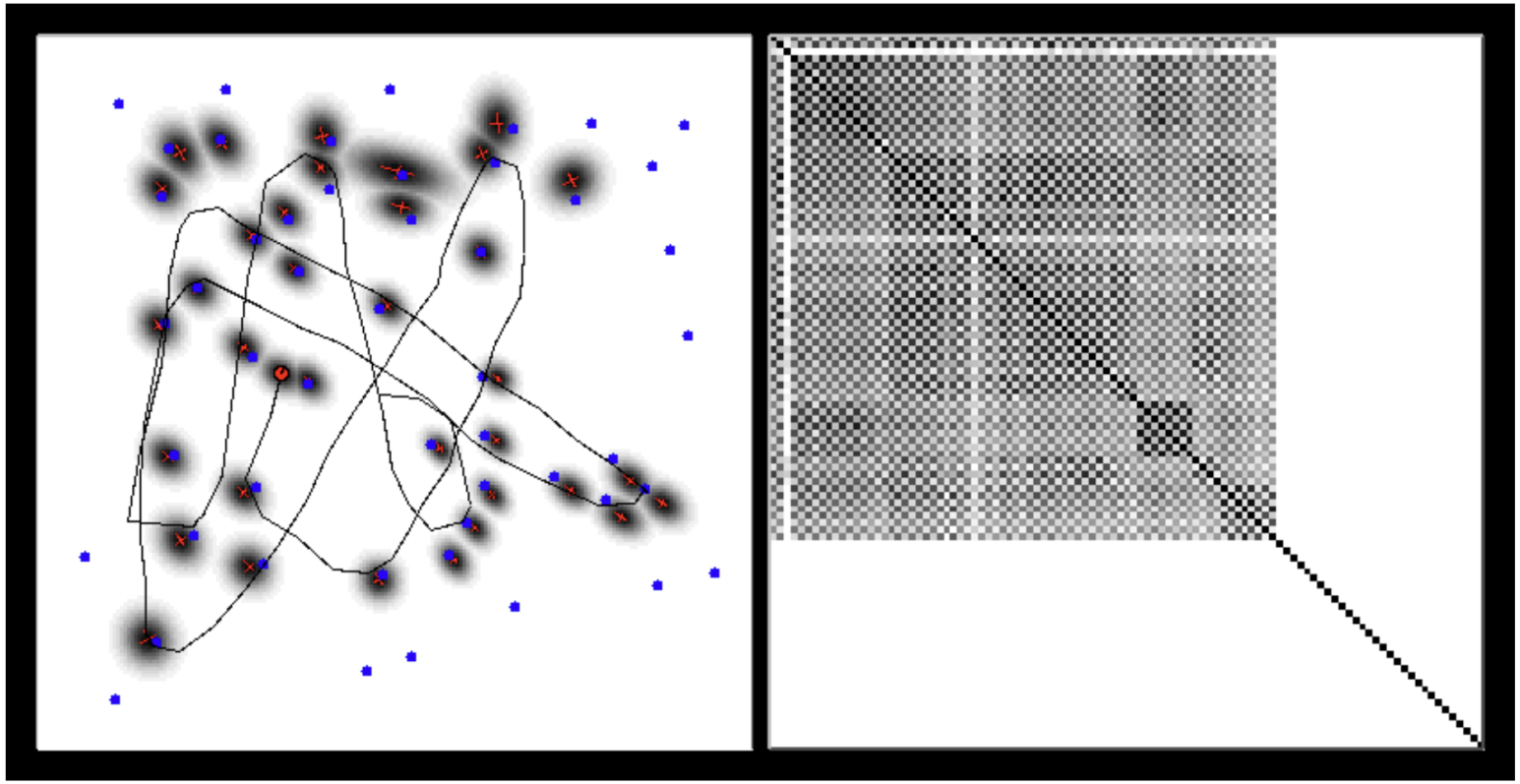
- Correlations can be observed by the state of the correlation matrix Σ:



Map | Correlation Matrix

EKF-SLAM Correlations

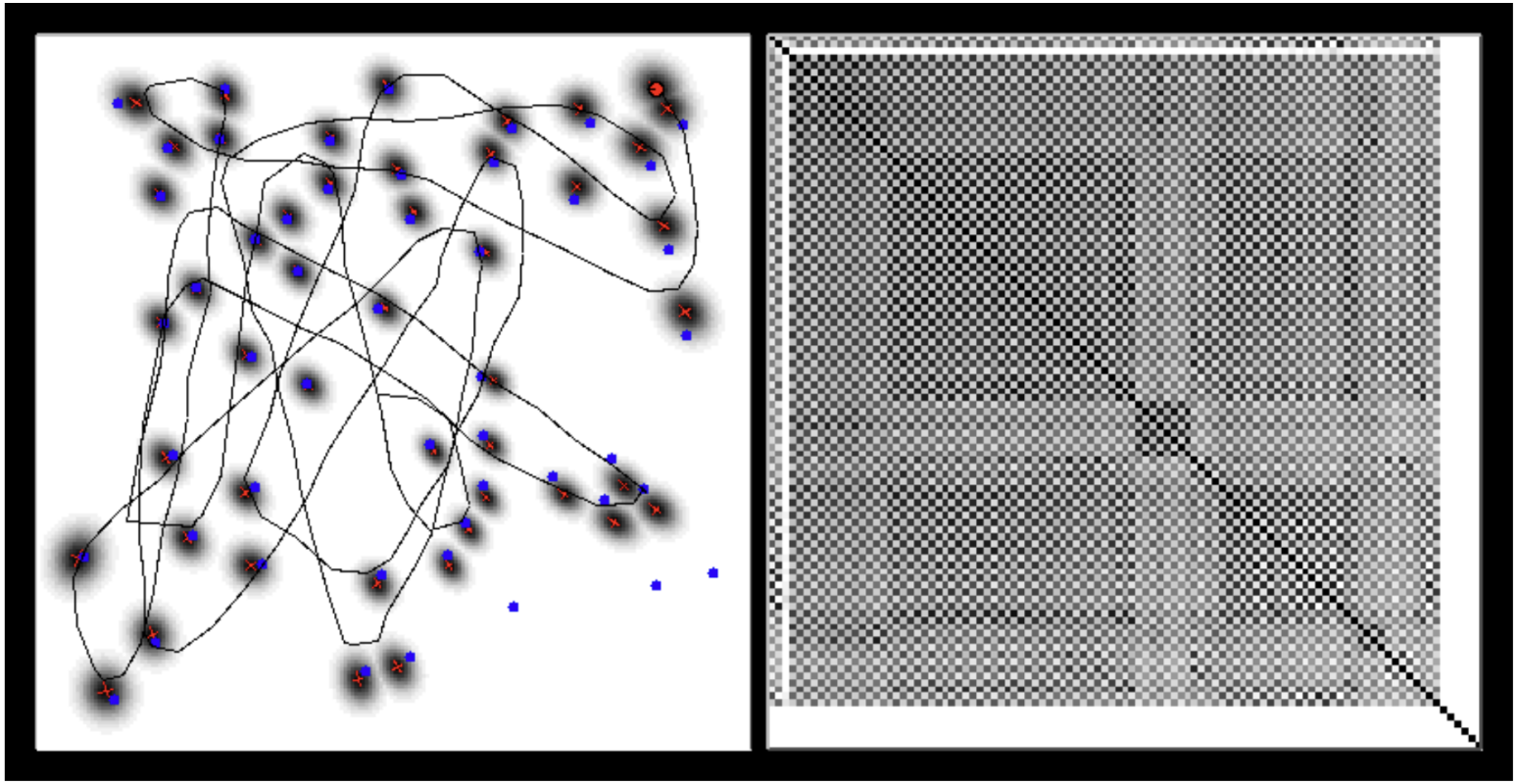
- Correlations can be observed by the state of the correlation matrix Σ:



Map | Correlation Matrix

EKF-SLAM Correlations

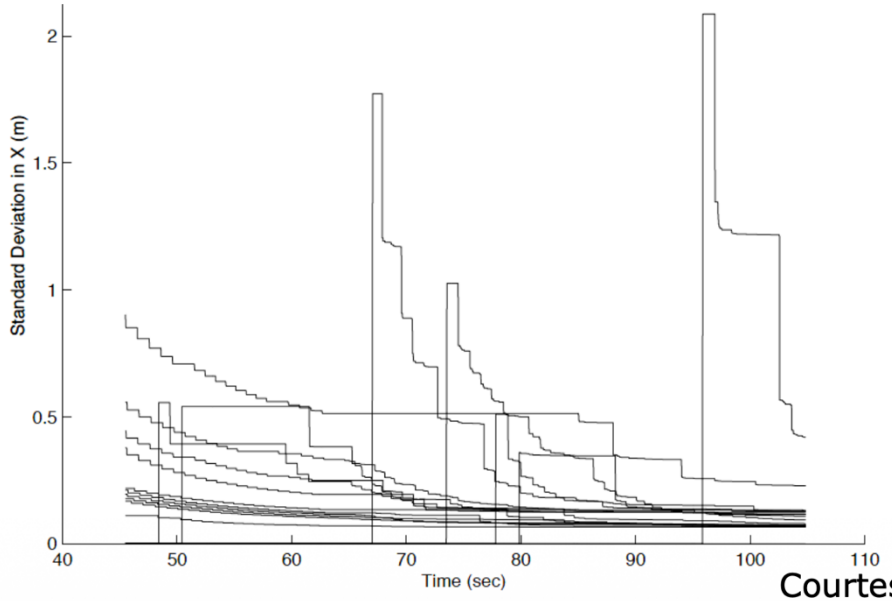
- Correlations can be observed by the state of the correlation matrix Σ:



Map | Correlation Matrix

EKF-SLAM Correlations

- The correlation between the robot's pose and the landmarks cannot be ignored.
- Assuming independence generates too optimistic estimates of the uncertainty.
- The determinant of any sub-matrix of the map covariance matrix decreases monotonically.
- New landmarks are initialized with maximum uncertainty.



Courtesy: Dissanayake

EKF-SLAM: Complexity

- Cubic complexity w.r.t. the measurement dimensionality
- Cost per step: dominated by the number of landmarks: $O(n^2)$
- Memory consumption: $O(n^2)$
- The EKF becomes computationally intractable for large maps!

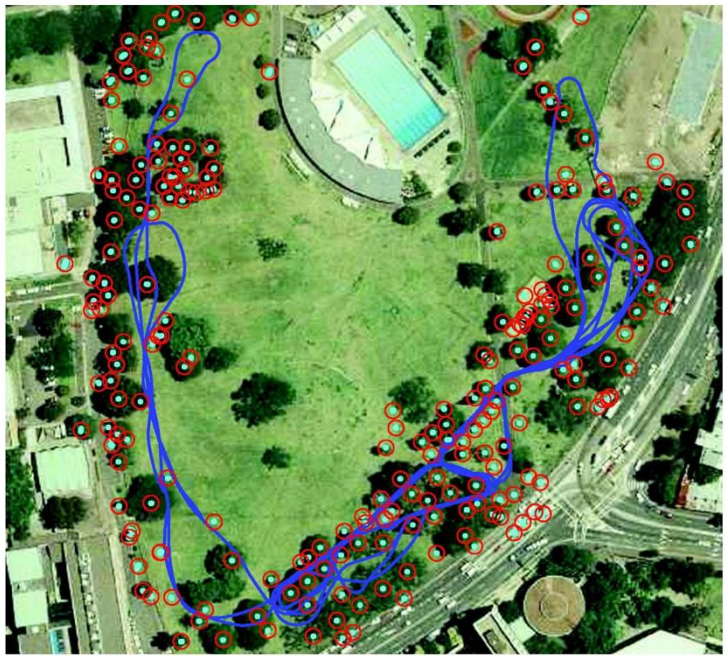
EKF-SLAM Summary

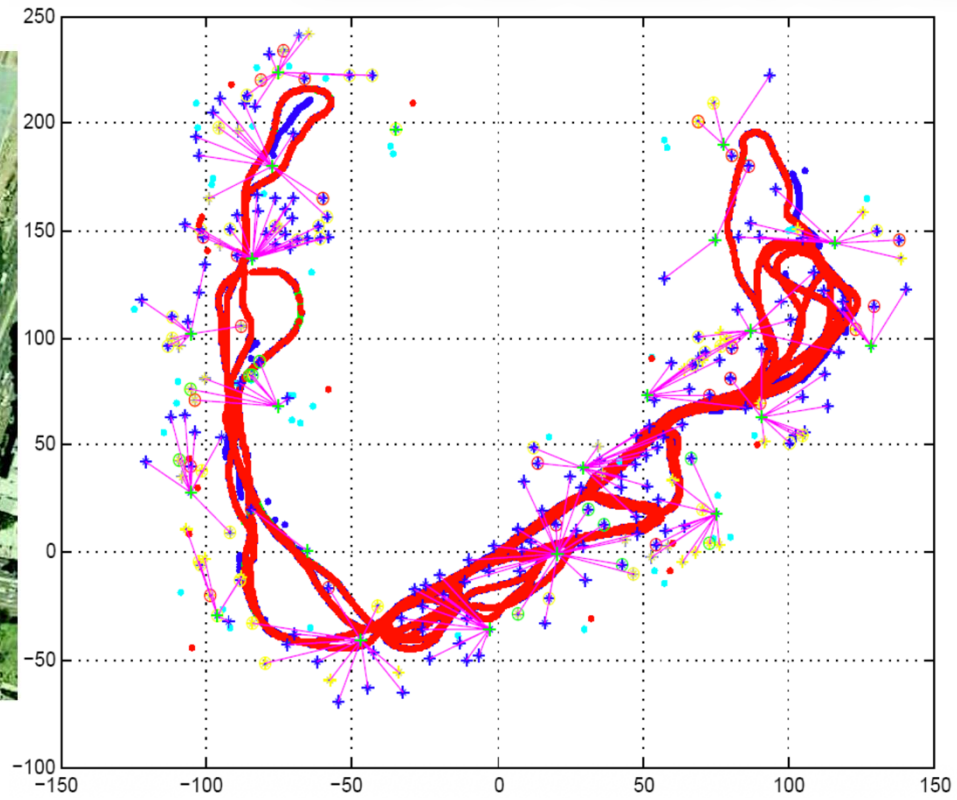
- Using the EKF to estimate pose and map in an SLAM fashion
- The first probabilistic SLAM approaches used the EKF
- Successful in medium-scale scenes
- Approximations exists to reduce the computational complexity
- Today mainly used for short-term estimates (VO)
- Unimodal (Gaussian) estimates only
- Convergence proof for the linear Gaussian case and then equivalent to least squares
- The smaller the noise the better the estimate in the non-linear case
- Can diverge if non-linearities are large

EKF-SLAM Examples

Example: Victoria Park Dataset



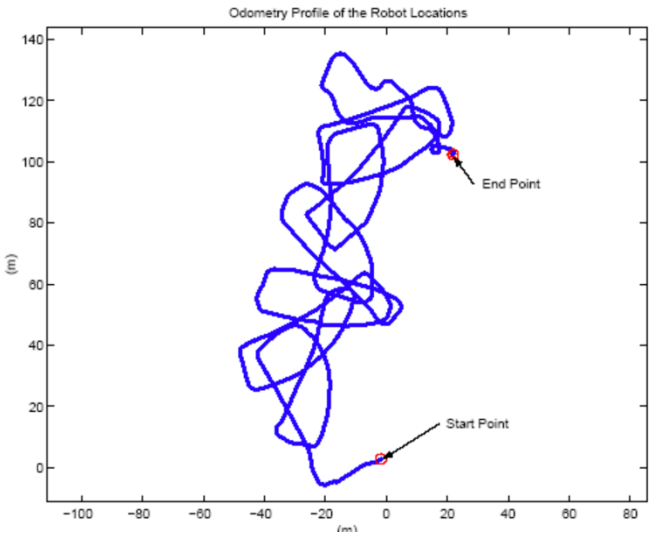




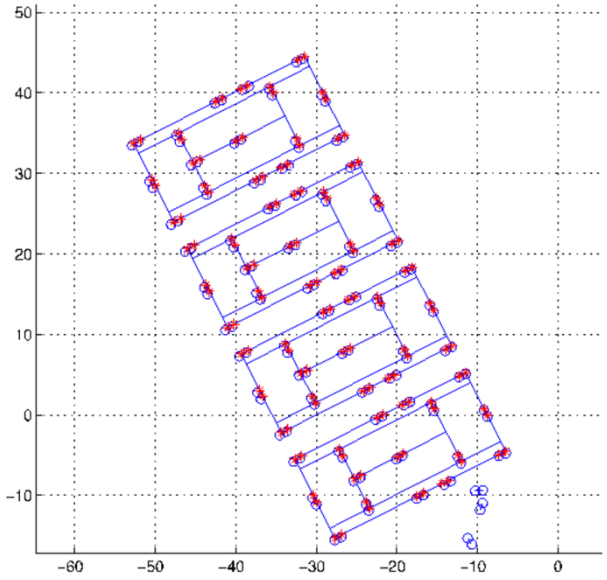
Courtesy: E. Nebot

Example: Tennis Court





Pure Odometry



SLAM Map

Sensor Networks and Sensor Fusion

- **Previous lectures:**
  - Mathematical details of Particle Filters
  - Joint Probability Data Association Filters
  - JPDAs and Particle Filters

- **This week:**
  - Simultaneous Localization and Mapping
    - Introduction
    - EKF SLAM
    - Loop Closure
    - Particle Filter / Rao-Blackwellized Particle Filter SLAM
    - "Hidden Robot" problem
- **After the Christmas Break:**
  - Bit more SLAM
  - (DNN-based) early fusion approaches.



Merry Robot Christmas!

Literature / Links

1. **S.Thrun &al., *Probabilistic Robotics*, https://robots.stanford.edu/probabilistic-robotics/**



Technische Hochschule Ingolstadt

Fakultät Elektro- und Informationstechnik

Thank you very much!